TabICL-M: A Tabular Foundation Model that Reads Missingness as Provenance

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Code, weights and every result table: https://github.com/Sompote/TabICL-M

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Abstract

Engineering databases are merged from laboratories, sites and campaigns, each of which recorded its own subset of the features with its own calibration; their gaps are block-structured and informative about the source, not random. Tabular foundation models treat missing values cell by cell, and we show that this is where they all lose: when the test rows come from a source the context has never seen, TabICLv2, TabPFN 2.5, TabPFN-3, classical imputers and gradient-boosted trees give up 0.02–0.08 AUC or 1–3 RMSE beyond what the gaps cost, and none recovers it. TabICL-M continues TabICLv2 on one observation: rows that share a missingness pattern come from the same source, so the pattern is free, label-free provenance. It encodes each value relative to its own pattern group, represents each row from its observed features only, carries a pattern token into in-context learning, and is trained with a source-structured missingness prior, block reconstruction and an offset-consistency loss; every addition is zero-initialised, so on complete data it equals TabICLv2. On 20 datasets with injected block missingness (5 seeds, held-out-source and random splits) TabICL-M is the best model of any kind when a held-out source carries an offset (mean rank 2.22 against 2.45 for TabPFN-3; 14 of 20 datasets; Elo 1325 against 1267 under TabArena's scoring), level with TabPFN-3 on classification, and behind it only on plain regression, at half the parameters and a third of the inference time. Every negative result on the way is reported, so that the remaining gap is attributed to the base regressor, not to the method.

1 Introduction

A foundation model for tabular data is pre-trained once on synthetic tables and then applied to a new table by in-context learning: the labelled rows are the context, the unlabelled rows are the queries, and no gradient step is taken (Hollmann et al., 2023, 2025; Qu et al., 2025). These models now lead the public leaderboards on small and medium tables (Erickson et al., 2025). They are trained, however, on complete synthetic tables, and the missing values they meet in practice are treated as a nuisance to be smoothed away, by imputation in the preprocessing or by a cell-wise missingness indicator in the encoder (Hollmann et al., 2025).

The tables that motivate this work look different. A geotechnical compaction database, the case that started the project, is a union of records from several laboratories; each laboratory measured the properties it had equipment for, and the same physical quantity measured in two laboratories differs by an offset. The result is a table in which the missingness is *block-structured* (a laboratory lacks whole columns, not scattered cells) and *informative about the source* (which columns a row lacks tells you where it came from), and in which the features carry a per-source shift. A model that imputes the column mean into a laboratory's

unmeasured columns, and treats the shifted values of another laboratory as if they were on a common scale, is biased in a way that no amount of context can correct.

Before building anything we asked where the headroom is. We took complete public tables, split their rows into synthetic sources, gave each source its own feature subset and optionally its own offset, and evaluated the released TabICLv2 under two splits: a random split, in which every source appears in the context, and a leave-one-source-out split, in which the test rows come from a source the context has never seen. The finding (Section 3) is sharp. Under the random split the source offset costs nothing: in-context learning calibrates it away, and mean imputation inside an in-context learner is already close to the information limit, so that no method, including TabPFN 2.5 and TabPFN-3, is separable from any other. Under the held-out source the same offset costs 0.02–0.08 AUC or 1–3 RMSE on every dataset, on top of the loss from the gaps, and every existing method loses it: imputers lose more than mean imputation, trees more still, and TabPFN as much as TabICL. That loss is a nuisance offset on values that are all present, not lost information, and it is the case a merged database presents every time a new source arrives.

TabICL-M targets exactly that case. Its principle is that the missingness pattern of a row is its provenance. Section 4 turns this into three additions to the three stages of TabICLv2 and two training objectives, all of which are zero-initialised or identity on complete data so that the model starts, and on complete tables stays, exactly at the released TabICLv2. Section 5 describes the continued pre-training on synthetic data only, Section 6 the benchmark against the released model, classical imputers, gradient-boosted trees and four versions of TabPFN, the ablation of each part on both tasks, and the negative results, and Section 7 what the whole says about where the remaining gap to TabPFN-3 lies.

Contributions. (i) A headroom study showing that the only regime in which current tabular foundation models leave a recoverable gap on incomplete data is a held-out source with its own feature subset and measurement offset. (ii) A source-aware architecture that reads a row's missingness pattern as provenance, with a source-relative value encoding, observed-only row attention and a pattern token, plus a block-structured prior and two objectives that train it, all bit-exact to TabICLv2 on complete data. (iii) The first tabular foundation model that beats TabPFN-3 when a new source carries an offset, at half its size and a third of its inference time, while remaining level with it on classification. (iv) A complete record of what did not work, so that the plain regression gap is attributed correctly to the base regressor.

2 Related work

Tabular foundation models. TabPFN (Hollmann et al., 2023) introduced in-context learning on synthetic tables drawn from structural causal models; TabPFN v2 (Hollmann et al., 2025) extended it to regression and larger tables, injected cell-wise missingness into its prior and added a missing indicator to its encoder. TabPFN 2.5 and TabPFN-3 are the current releases of that line; we compare against all of them. TabICL (Qu et al., 2025) and TabICLv2 (Qu et al., 2026) factor the model into a column-wise set transformer (Lee et al., 2019), a row-wise transformer and a dataset-wise in-context learner, which scales to hundreds of thousands of rows. TabICL-M is a fork of TabICLv2 and inherits that structure.

Missing values in deep tabular models. NAIM (Caruso et al., 2024) masks missing features out of attention instead of imputing them, an idea our observed-only row attention shares, inside an in-context learner. ReMasker (Du et al., 2024) and VIME (Yoon et al., 2020) train tabular models with masked-cell reconstruction; our reconstruction objective hides blocks of columns of a pseudo-source rather than random cells. All three treat missingness as cell-wise and random; none exploits the pattern as a source identifier. Classical

imputation, multiple imputation by chained equations in particular (van Buuren and Groothuis-Oudshoorn, 2011), is the standard preprocessing; we show it is worse than mean imputation in front of an in-context learner. The distinction between missing completely at random, at random and not at random goes back to Rubin (1976); block-structured, source-informative missingness fits none of the three.

Source shift. Normalising per domain to remove a nuisance shift is an old idea in vision (Li et al., 2016). Our source-relative value encoding is its tabular, label-free analogue, with the twist that the domain is inferred from the missingness pattern and the normalised value is added as an input rather than replacing the raw one, so that real between-source signal is not thrown away.

Benchmarks. TabArena (Erickson et al., 2025) is a living benchmark of tabular models on curated i.i.d. datasets; BeyondArena (Purucker et al., 2026) adds temporal and grouped splits. Their datasets are complete, so they cannot show the effect we target; we report results on them separately in the repository.

3 Where the headroom is

We start from six complete public tables (breast cancer, wine, diabetes, credit-g, adult, Boston housing), delete cells under a stated mechanism at a stated rate, and compare models that handle the gaps in different ways. The *block* mechanism assigns rows to two to five sources and removes a different subset of columns from each; *block_shift* additionally gives every source an additive offset $\mathcal{N}(0, 0.5\sigma_j)$ and extra noise on each numeric feature j, imitating laboratories that measure differently. Two splits are used: a stratified random split, and a leave-one-source-out split in which one whole synthetic source is the test set. Five seeds, rates 0.3 and 0.5.

Table 1: The released TabICLv2 (mean imputation) at 50 % missingness. Source shift costs nothing under a random split and 0.02–0.08 AUC or 1–3 RMSE under a held-out source. Mean over 5 seeds.

dataset	complete	block, random	block, held-out	**block_shift, held-out**	shift-specific loss
adult (AUC)	0.912	0.865	0.847	**0.800**	-0.047
credit-g (AUC)	0.814	0.722	0.704	**0.683**	-0.021
wine (AUC)	1.000	0.996	0.990	**0.914**	-0.076
Boston (RMSE)	2.71	4.84	5.84	**7.08**	$+1.24$
diabetes (RMSE)	55.8	65.3	66.4	**69.1**	$+2.7$

Table 1 gives the base model's numbers. Three facts follow. First, under the random split the offset is harmless, because every source is represented in the context and in-context learning absorbs a constant per-source shift on its own. Second, under the held-out source the offset costs every dataset a further 0.02–0.08 AUC or 1–3 RMSE. That loss is recoverable in principle: the values are all present, only their scale is unknown. Third, every alternative loses more. Iterative imputation in front of the same model is worse than mean imputation on all six datasets under the held-out source (adult 0.677, wine -0.042 AUC, Boston $+1.35$ RMSE); k-nearest neighbour imputation is worse still; XGBoost and CatBoost drop to 0.79 and 0.78 AUC on adult and 8.6 and 9.5 RMSE on Boston; and TabPFN v2 (0.816), TabPFN 2.5 (0.822) and TabPFN-3 (0.812) sit in the same band as TabICLv2 on adult. The first-generation TabICL-M checkpoint, which adds only value-level parts (a missing indicator, an absence vector and cell-wise reconstruction), moved these numbers by less than the seed spread. A fixed pattern-conditional normalisation recovers part of the shift loss (adult $+0.025$ AUC) but hurts when there is no offset (credit-g -0.012), which is the argument for learning the encoding instead of hard-coding it.

4 Method

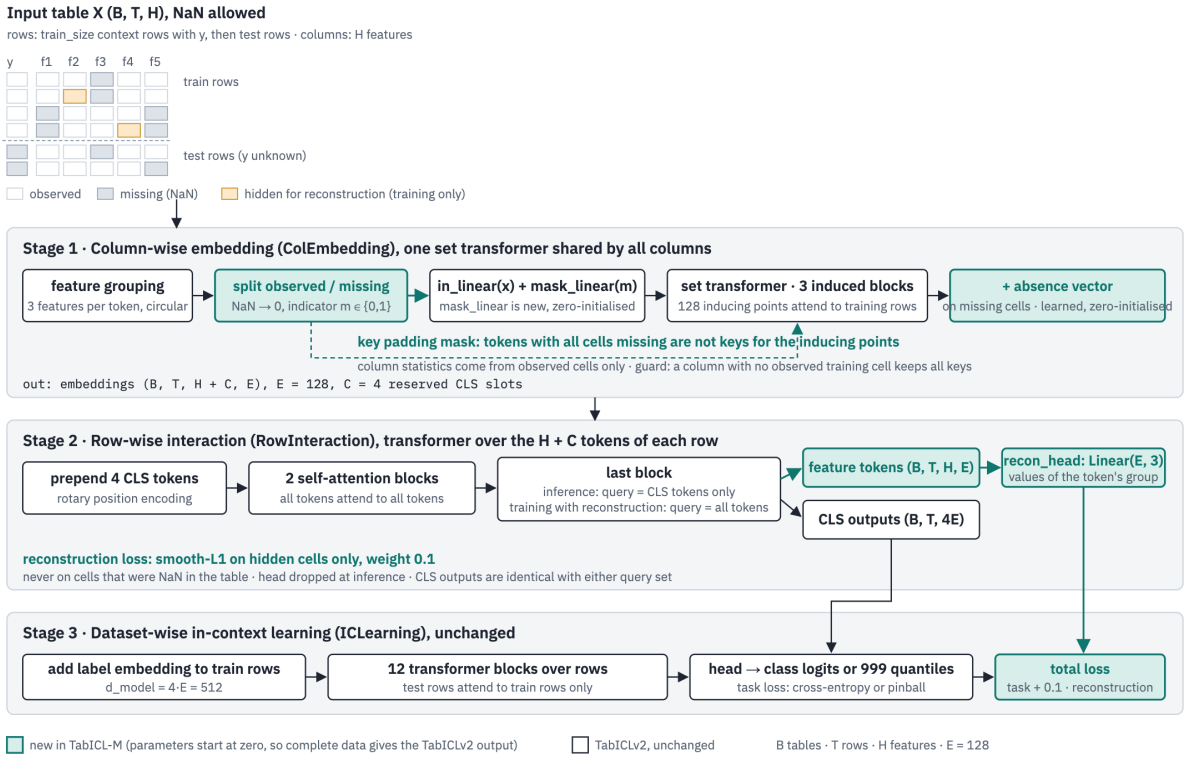

Figure 1: The three stages of TabICLv2 with the additions of TabICL-M in green. Every new parameter starts at zero, so on a table with no missing cell the output is identical to TabICLv2.

4.1 The principle

In a table merged from several sources, two rows that lack exactly the same columns almost always come from the same source. The binary missingness pattern of a row is therefore a source identifier that costs nothing, needs no provenance column and survives any preprocessing that keeps NaN. TabICL-M uses it at every stage (Figure 1). We write a table as $X \in \mathbb{R}^{T \times H}$ with T rows, H features and NaN at missing cells, $m_{tj} = \mathbb{1}[X_{tj} \text{ missing}]$, and $p(t) = (m_{t1}, \ldots, m_{tH})$ for the pattern of row t; the first n rows carry a label.

4.2 Stage 1: source-relative value encoding

TabICLv2 embeds each column with a set transformer whose 128 inducing points attend to the labelled rows, so the column statistics that scale a value come from the whole context. TabICL-M groups rows by pattern, $G(t) = \{t' : p(t') = p(t)\}$, and feeds every observed cell twice: as its raw value and as

$$\tilde{x}_{tj} = \frac{x_{tj} - \mu_j(G(t))}{\sigma_j(G(t))}, \tag{1}$$

the value standardised within the rows of its own pattern group (patterns with fewer than eight rows fall back to the table-wide statistics). Labelled and unlabelled rows are pooled for these statistics, and no label

is used, so a source that appears only in the test rows is centred on itself. The two values enter the token through separate linear maps, $\mathrm{in}(x) + \mathrm{rel}(\tilde{x}) + \mathrm{mask}(m)$, and the map rel is zero-initialised. Because the raw value is still present, the model can learn how much of a between-source difference is nuisance and how much is signal, instead of having that decided by a fixed normalisation. Missing cells are zero-filled, hidden from the keys of the inducing-point attention so that statistics come from observed cells only, and receive a learned absence vector on the output.

4.3 Stage 2: observed-only rows and a pattern token

The row-wise transformer of TabICLv2 attends over the H feature tokens of a row plus four CLS tokens whose outputs, concatenated, form the row representation. TabICL-M makes two changes. Feature tokens whose cells are all missing are excluded from the attention keys, so the representation of a row is a function of its observed features only; a test row with a combination of observed features never seen jointly in the context is then represented by the same mechanism as any context row, and the in-context stage can reason through the overlaps. Second, a learned query reads out the row's token set once, with fixed sinusoidal position codes so that it knows *which* features are present, and its output is added to the row representation through a zero-initialised projection. This pattern token carries the source identity into the in-context learner, which can then weight context rows by how comparable they are to a query. The remaining stage, twelve transformer blocks over rows with test rows attending to labelled rows, is unchanged.

4.4 Objectives

Two losses are added to the task loss (cross-entropy, or a pinball loss over 999 quantiles for regression). *Block reconstruction* hides a block of cells, a random subset of rows losing a random subset of columns, i.e. a pseudo-source losing part of its features, and reconstructs them from the per-feature token outputs of the row-wise stage through a small linear head with a smooth-ℓ_1 loss; the head is dropped at inference. Reconstructing a block, unlike reconstructing random cells, means completing a source's absent features from the other sources. *Offset consistency* builds a second view of each incomplete table in which every pattern group receives its own additive offset and noise on the numeric features and pulls the prediction on that view towards the (detached) prediction on the clean view, by KL divergence for classification and squared error for regression. It trains invariance to the nuisance directly rather than hoping the prior induces it.

4.5 Why the parts and not others

The alternatives we did not take are informative. A missingness indicator alone (TabPFN's route, and our first checkpoint) cannot represent *which* source a row is from; the ablation in Section 6.4 shows the pattern token, which can, is the single most valuable part for classification. Hard-coding the normalisation helps under shift and hurts without it; learning it through a zero-initialised map avoids the trade-off. Imputing before the model destroys the pattern; excluding absent tokens from attention keeps it. A larger model or a different prior alone was ruled out by the headroom study: TabPFN 2.5, with a third of our parameters, and TabPFN-3, with twice as many, lose the same amount under a held-out source, so capacity is not what is missing. Finally, every addition is zero-initialised or identity on complete data, which lets the run start exactly at the released weights and guarantees that the published behaviour of TabICLv2 on complete tables is preserved; the tests enforce this to 10^{-6}.

5 Training

Data. TabICL-M is trained on synthetic tables only. TabICLv2 pre-trains on tables generated from structural causal models with random graphs, activations and noise, standardised per table (Qu et al., 2026). TabICL-M applies a missingness transform to every generated table (Figure 2). Rows are split into 2 to 8 sources with Dirichlet row shares; each source observes a fraction 0.2 to 1 of the features, with a core set seen by all; numeric features receive a per-source offset of up to 0.5σ (1.2σ in the second stage) and noise of up to 0.3σ (0.5σ); with probability 0.75 (0.9) the sources occupy contiguous row blocks so that some fall entirely in the test part, the leave-one-source-out case; with probability 0.5 the source id is appended as a categorical column. Cell-wise gaps under MCAR, MAR and MNAR mechanisms, including detection-limit censoring, are layered on top, each driven to a sampled rate by bisection. Two safety rules keep at least two observed labelled cells per feature and one observed feature per row; the target is never masked. No real dataset is used for training, and none of the evaluation datasets is seen.

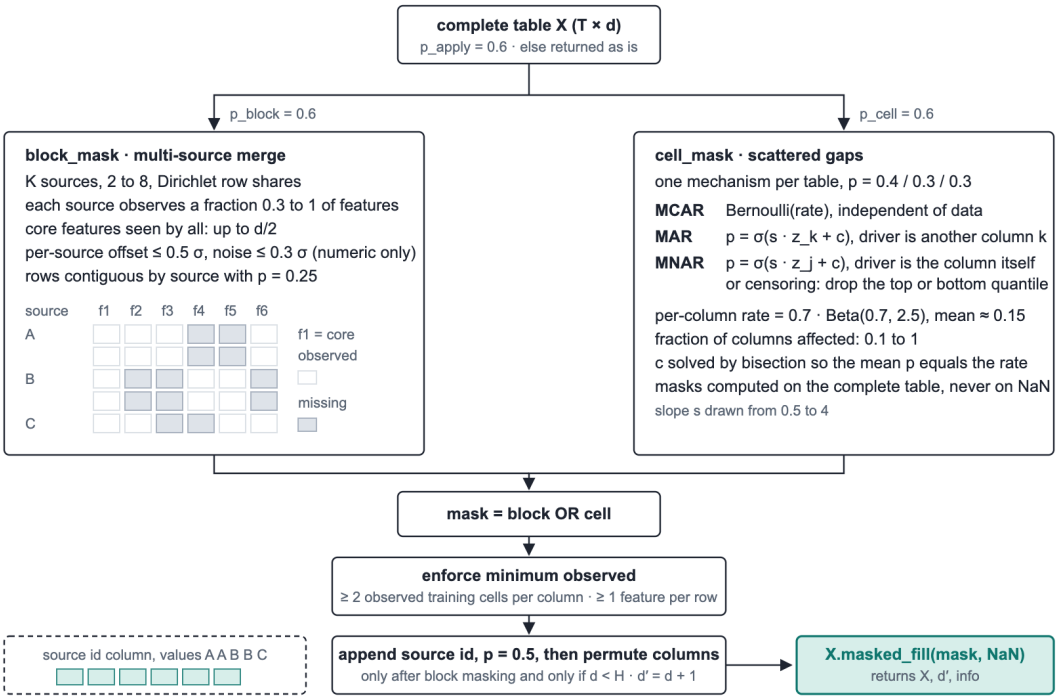

Figure 2: How a complete synthetic table becomes an incomplete one: block masking by source, cell-wise masking, their union, the safety rules and the optional source-id column.

Recipe. Training continues from the released TabICLv2 classifier and regressor. Stage 4 runs 10 000 steps at batch 32 with sequences of 400 to 8 192 rows and up to 100 features, Muon optimiser (Jordan et al., 2024), cosine schedule, learning rate 5×10^{-5}, block reconstruction with weight 0.1, offset consistency with weight 0.1 on a quarter of the micro-batches. Stage 4b continues for another 10 000 steps at 3×10^{-5} on the stronger source effects given above, because the stage-4 model's consistency loss had fallen to 0.005, showing it was invariant to the shifts it saw but not yet to the larger ones of the evaluation. Each stage takes about 9.5 hours per task on one NVIDIA RTX 5090 in bfloat16 (2.7–3.3 s per step); the second view of the consistency loss adds about 15 %. The classifier's task loss fell from 0.60 to 0.57 while the reconstruction

loss fell from 0.21 to 0.19 and the consistency loss stayed near 0.005; the regressor's pinball loss stayed near 0.08. The classifier has 27.7 M parameters and the regressor 28.7 M, against 27.6 M and 28.5 M for TabICLv2: the additions cost 0.1–0.2 M.

6 Experiments

Protocol. Twenty public datasets (13 classification, 7 regression; from scikit-learn and OpenML, subsampled to at most 3 000 rows), block and block_shift missingness at rates 0.3 and 0.5, five seeds, both splits of Section 3: about 400 conditions per split. Models: the released TabICLv2 with mean imputation ("TabICLv2"), TabICL-M, TabPFN 2.5 and TabPFN-3 at their default settings and public checkpoints, and CatBoost. All in-context learners use eight ensemble members. We report the mean rank over the five models (1 = best), paired win/loss counts against TabPFN-3 on identical splits and deleted cells, and the number of datasets on which the per-dataset mean is better. Everything is reproducible from the repository (`results/broad`); the datasets are listed in Appendix A and the per-dataset numbers in Appendix B.

Table 2: Mean rank over five models (1 = best) on the 20-dataset benchmark; paired counts and datasets won are TabICL-M against TabPFN-3.

task	split, condition	**TabICL-M**	TabPFN-3	TabPFN 2.5	TabICLv2	CatBoost	vs TabPFN-3
Classification, 13 datasets							
	held-out source, with offset	**2.35**	2.41	2.48	3.08	4.68	61/69, 9 of 13
	held-out source, all	2.36	2.36	2.58	3.01	4.69	123/133, 9 of 13
	held-out source, no offset	2.37	**2.31**	2.68	2.94	4.70	62/64, 7 of 13
	random split	**2.35**	2.45	2.69	2.77	4.74	131/120, 8 of 13
Regression, 7 datasets							
	held-out source, with offset	**1.97**	2.53	2.63	3.44	4.43	**44/26, 5 of 7**
	held-out source, all	2.24	**2.14**	2.61	3.41	4.61	67/73, 3 of 7
	held-out source, no offset	2.50	**1.74**	2.59	3.37	4.80	23/47, 2 of 7
	random split	2.79	**1.79**	2.75	3.00	4.67	42/98, 1 of 7
Both tasks, 20 datasets							
	held-out source, with offset	**2.22**	2.45	2.53	3.21	4.59	**105/95, 14 of 20**
	held-out source, all	2.32	**2.28**	2.59	3.15	4.66	190/206, 12 of 20
	random split	2.51	**2.22**	2.71	2.85	4.71	173/218, 9 of 20

6.1 Main results

Table 2 is the result. When a held-out source carries a measurement offset, TabICL-M ranks first on both tasks and pooled (2.22 against 2.45 for TabPFN-3 and 2.53 for TabPFN 2.5) and wins 14 of 20 datasets against TabPFN-3; on regression the margin is clear (44 wins to 26, 2.1 % lower RMSE, 5 of 7 datasets). On classification it is level with TabPFN-3 everywhere: identical rank over all held-out-source conditions, a touch ahead on random splits and under offset, and the differences are 0.002–0.005 AUC, inside seed noise. On regression without an offset TabPFN-3 leads clearly (23/47 under a held-out source, 42/98 on random splits). Against every other model TabICL-M wins every summary on both tasks: 219/177 and 16 of 20 datasets against TabPFN 2.5, 281/116 and 18 of 20 against the released TabICLv2 under held-out sources. Against its own base the improvement is largest where it was designed to be: on regression under a held-out source it lowers RMSE by 8.5 % with offset (59 wins, 11 losses, 7 of 7 datasets; diamonds 2935 to 2208, diabetes 69.1 to 65.1) and 6.1 % without, while on complete data the two are within 0.7 % (17 wins, 18

losses). Under a held-out source the regressor is also better calibrated: its 80 % interval covers 0.81 with a width of 1.6 target standard deviations, where TabICLv2 over-covers at 0.87 with width 2.1.

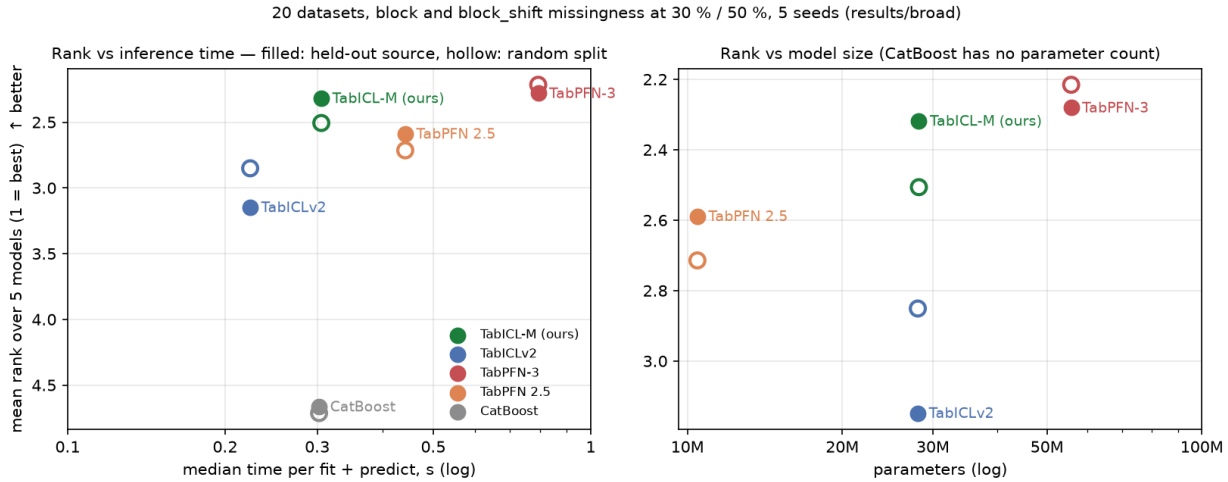

Figure 3: Mean rank on all incomplete conditions of the benchmark against median inference time (fit plus predict, one RTX 5090, eight ensemble members) and against parameter count. Filled: held-out source; hollow: random split. TabICL-M matches TabPFN-3 with half the parameters and a third of the time and improves on TabICLv2, equal in size and speed, by 0.8 rank points on held-out sources.

6.2 Cost

Figure 3 places accuracy against cost. TabICL-M (28 M parameters, 0.3 s per fit and predict) matches TabPFN-3 (56 M, 0.8 s) in rank and is the most accurate model per parameter and per second among the foundation models; TabPFN 2.5 is smaller (10.5 M) but ranks behind both. Because the additions cost 0.2 M parameters and one extra read-out, TabICL-M has the size and speed of TabICLv2 while being 0.8 rank points better on held-out sources.

6.3 TabArena-style scoring

TabArena (Erickson et al., 2025) scores methods by Elo from pairwise comparisons per task, with bootstrap confidence intervals, alongside rank, win rate and improvability. We apply its evaluator to the same runs, treating each (dataset, mechanism, rate) as a task and each seed as a repeat, with log loss for classification and RMSE for regression as the error and Elo anchored to TabICLv2 at 1000 (Figure 4, Table 3). On held-out sources TabICL-M leads TabPFN-3 by 58 Elo (1325 against 1267; win rate 0.80 against 0.73; mean reciprocal rank 0.70 against 0.61) at a third of its inference time; on random splits TabPFN-3 leads by 38 Elo with confidence intervals of several hundred points, a tie. CatBoost trails both splits by more than 400 Elo.

6.4 Ablation

Each part has a switch. Table 4 trains one variant per switch for 3 000 steps on the source-aware prior and evaluates it on the held-out-source-with-offset condition; for the classifier the metric is the win rate against mean imputation, for the regressor the mean RMSE gain over the released regressor.

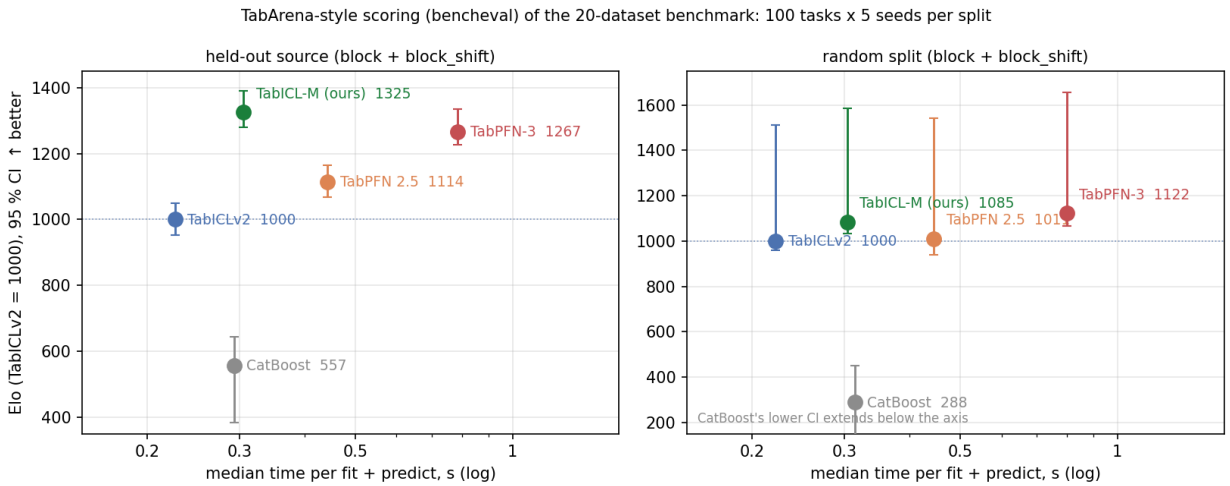

Figure 4: TabArena-style scoring of the benchmark: Elo with 95 % bootstrap confidence intervals, anchored to TabICLv2 at 1000, against median inference time (fit plus predict, one RTX 5090). 100 tasks × 5 seeds per split. CatBoost's lower interval on the random split extends below the axis.

Table 3: TabArena-style leaderboard of the benchmark (bencheval). Elo anchored to TabICLv2 = 1000; improvability is the mean relative distance to the best method per task.

	held-out source			random split		
model	Elo (95 % CI)	rank	win rate	Elo (95 % CI)	rank	win rate
TabICL-M (ours)	**1325** (+65/ − 45)	**1.79**	**0.80**	1085 (+500/ − 51)	2.34	0.67
TabPFN-3	1267 (+68/ − 40)	2.07	0.73	**1122** (+533/ − 54)	**2.13**	**0.72**
TabPFN 2.5	1114 (+51/ − 46)	2.86	0.54	1011 (+533/ − 70)	2.76	0.56
TabICLv2	1000 (+50/ − 47)	3.42	0.40	1000 (+512/ − 42)	2.82	0.55
CatBoost	557 (+87/ − 175)	4.86	0.04	288 (+162/ − 2005)	4.95	0.01

Table 4: Ablation on held-out sources with offset. Classifier: win rate against mean imputation (4 datasets). Regressor: mean RMSE gain over the released regressor (7 datasets). One switch off at a time, 3 000 steps each.

variant	classifier win rate	regressor RMSE gain
all parts on	0.73	+5.1 %
without source-relative values	0.70	+4.9 %
without observed-only rows	0.68	+4.8 %
without pattern token	0.65	+5.3 %
without block reconstruction and consistency	0.70	+5.1 %
all parts off (new prior only)	0.62	+4.5 %
first checkpoint (old prior, value-level parts)	0.62	—
all parts on, full training (10k / 20k steps)	0.77	+8.6 %

For the classifier every part helps, the pattern token and the observed-only rows most, and the new prior alone adds nothing over the first checkpoint: the gain is in the architecture. For the regressor the picture differs in one respect: the prior alone already gives +4.5 %, the parts add about one point at 3 000 steps, and training length adds the rest, to +8.6 % at 20 000 steps, the only variant ahead of TabPFN-3 there. No part hurts either task on complete data (every regression variant is within +0.2 to +1.0 % of the released regressor), so the parts are safe to keep on everywhere.

6.5 What did not close the regression gap

The one place TabPFN-3 leads is regression without a source offset. We probed it exhaustively, on kin8nm, space_ga and diamonds where the gap is largest (5–15 % on complete data), and report the results because they locate the cause. *Test-time levers*: 32 ensemble members, the median instead of the mean, a target power-transform ensemble and an extra quantile-normalisation member improve our own ranking slightly and leave the head-to-head unchanged (41 wins, 97 losses on random splits); TabPFN-3 gains nothing from 32 members either. *Weight soup* of the 10k and 20k regressors: $+0.08\,\%$. *Continued training* on a prior with half the tables complete: 85 wins, 90 losses against the 20k model after 3 500 steps, stopped. *Test-time fine-tuning* on the context rows recovers a third of the kin8nm gap (0.0761 to 0.0736 against 0.0721) and half of space_ga's, then plateaus at 100 epochs; TabPFN-3 fine-tuned the same way on the same rows gets slightly worse, so its zero-shot model stays best. *A TabPFN-style histogram head* (1 000 per-table equal-mass buckets, log-density loss, output layer from zero, 20 000 steps) is worse than the quantile head on every regression dataset on complete data (9 wins, 26 losses, 7 % higher RMSE). *A sample-efficiency curve* shows the kin8nm gap at every training size, narrowing from 13 % at 300 rows to 5 % at 2 100, so it is not a limitation of in-context learning from little data; and TabPFN 2.5, with 10 M parameters, beats us there too, so it is not capacity. What remains is the quality of the base regressor on smooth nonlinear functions, inherited from TabICLv2 (whose released regressor ranks 6.4 against TabPFN-3's 4.0 on these datasets under a held-out source) and not touched by missingness training, adaptation on the context, or the output distribution.

7 Discussion

Where the method pays and why. The gain of TabICL-M is confined to the regime it was built for, and that is the honest reading of Table 2. Under a random split every source is in the context and there is nothing for a source-aware mechanism to fix; under a held-out source without offset the missing block is a loss of information that no representation can undo; only under a held-out source with an offset is there a nuisance to remove, and there TabICL-M removes more of it than any other model. The ablation supports the mechanism: the pattern token, which carries the source identity, and the observed-only rows, which let a row from a new feature combination be represented at all, are the two parts that matter most for classification, and the first-generation checkpoint, which could see *that* a cell was missing but not *which source* a row came from, gained nothing.

What a fair comparison required. Three design choices decided what could be claimed. Pairing every comparison on identical splits and deleted cells turned seed-level noise into countable wins and losses. Splitting by task showed that the pooled second place behind TabPFN-3 (2.32 against 2.28) is entirely regression without a source offset, with classification level. And probing that deficit with every lever we could think of (Section 6.5) placed it in the base regressor rather than in the method. Without any one of these the paper would have claimed either too much or the wrong thing.

Limitations. All source structure in the benchmark is synthetic, imposed on complete public tables; the leave-one-laboratory-out test on the compaction database that motivated the work has not been run. Five seeds give paired win rates of 52–57 % under offset against TabPFN-3, a rank difference rather than a significant margin per dataset. TabArena and BeyondArena, whose datasets are complete, cannot show the effect we target; our TabArena-Lite and BeyondArena grouped-split runs are in progress and will be reported in the repository, with the expectation of a TabICLv2-level placement on TabArena. Finally, one RTX 5090

and 20 000 continuation steps are a small budget against the pre-training of TabPFN-3, and we do not claim to have exhausted what a larger one would give the base regressor.

Reproducibility. The repository contains the weights (git LFS), the prior, the training scripts with every switch used here, the evaluation runner, the raw per-fit results of every experiment in this paper (`results/`), the scripts that produce each table and figure from them, and the pipelines that ran the training and evaluation unattended. Seeds, splits and deleted cells are deterministic functions of the seed, so every paired comparison can be recomputed. The hyperparameters are in Appendix C.

8 Conclusion

Merged engineering databases are incomplete in a structured way: each source measured its own columns on its own scale. We showed that this is the one regime in which current tabular foundation models leave a recoverable gap, and that reading a row's missingness pattern as its provenance, through a source-relative value encoding, observed-only row attention and a pattern token trained on a source-structured prior, recovers it. TabICL-M is the best model of any kind when a new source arrives with its own calibration, level with TabPFN-3 on classification, and behind it only on plain regression, at half the parameters and a third of the inference time. The parts are zero-initialised, so nothing that TabICLv2 does on complete data is lost. The weights, the code, the benchmark and every negative result are public.

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A Datasets

Twenty public datasets, subsampled to at most 3 000 rows before the 70/30 split; train rows are the median over seeds and splits. Classification metric AUC (log loss for the TabArena-style scoring), regression metric RMSE.

Table 5: The 20 datasets of the benchmark.

dataset	source	train rows	features	task
adult	OpenML 1590	2100	14	classification
bank-marketing	OpenML 1461	2100	16	classification
breast cancer	scikit-learn	398	30	classification
churn	OpenML 40701	2100	20	classification
climate-model	OpenML 40994	378	18	classification
cmc	OpenML 23	1031	9	classification
credit-g	OpenML 31	700	20	classification
ilpd	OpenML 1480	408	10	classification
kc1	OpenML 1067	1476	21	classification
kc2	OpenML 1063	365	21	classification
pima-diabetes	OpenML 37	537	8	classification
qsar-biodeg	OpenML 1494	738	41	classification
wine	scikit-learn	124	13	classification
Boston housing	OpenML 531	354	13	regression
bodyfat	OpenML 560	176	14	regression
diabetes	scikit-learn	309	10	regression
diamonds	OpenML 42225	2100	9	regression
kin8nm	OpenML 189	2100	8	regression
openml-44970	OpenML 44970	635	6	regression
space_ga	OpenML 507	2100	6	regression

B Per-dataset results

Mean over 5 seeds at 50 % missingness; AUC for classification (higher is better), RMSE for regression (lower is better); best per row in bold.

Table 6: Held-out source with offset (block_shift, rate 0.5).

dataset	TabICL-M	TabPFN-3	TabPFN 2.5	TabICLv2	CatBoost
adult	0.815	0.812	**0.822**	0.800	0.783
bank-marketing	0.744	0.734	0.734	**0.748**	0.634
breast cancer	**0.985**	0.982	0.984	0.963	0.955
churn	**0.696**	0.668	0.682	0.651	0.635
climate-model	**0.773**	0.738	0.744	0.759	0.673
cmc	**0.655**	0.643	0.653	0.642	0.591
credit-g	**0.685**	0.673	0.679	0.683	0.624
ilpd	0.682	0.679	**0.703**	0.694	0.537
kc1	0.803	0.803	**0.803**	0.789	0.745
kc2	**0.869**	0.866	0.857	0.858	0.600
pima-diabetes	0.701	**0.708**	0.699	0.705	0.562
qsar-biodeg	**0.892**	0.887	0.886	0.879	0.817
wine	**0.941**	0.925	0.935	0.914	0.806
Boston housing (RMSE)	**6.716**	6.814	6.749	7.076	9.488
bodyfat (RMSE)	5.651	**5.359**	5.559	6.159	8.318
diabetes (RMSE)	**65.12**	65.45	65.56	69.10	78.81
diamonds (RMSE)	2207.69	**2162.97**	2339.77	2934.62	3028.90
kin8nm (RMSE)	**0.234**	0.235	0.238	0.239	0.278
openml-44970 (RMSE)	**1.346**	1.347	1.388	1.430	1.511
space_ga (RMSE)	**0.186**	0.199	0.197	0.204	0.235

Table 7: Held-out source without offset (block, rate 0.5).

dataset	TabICL-M	TabPFN-3	TabPFN 2.5	TabICLv2	CatBoost
adult	0.847	**0.849**	0.845	0.847	0.802
bank-marketing	**0.780**	0.769	0.777	0.775	0.693
breast cancer	0.989	**0.991**	0.991	0.968	0.972
churn	0.757	**0.767**	0.765	0.759	0.713
climate-model	**0.737**	0.719	0.714	0.726	0.652
cmc	0.660	0.664	0.661	**0.665**	0.591
credit-g	0.701	0.686	0.693	**0.704**	0.603
ilpd	0.705	0.707	**0.711**	0.696	0.628
kc1	**0.822**	0.819	0.800	0.679	0.777
kc2	0.875	**0.876**	0.872	0.868	0.680
pima-diabetes	0.817	**0.819**	0.817	0.816	0.775
qsar-biodeg	0.875	**0.899**	0.884	0.851	0.843
wine	**0.992**	0.991	0.991	0.990	0.958
Boston housing (RMSE)	5.675	**4.942**	5.862	5.840	8.031
bodyfat (RMSE)	5.044	**4.526**	4.888	5.346	7.599
diabetes (RMSE)	64.11	63.98	**63.65**	66.36	73.23
diamonds (RMSE)	1463.38	**1292.58**	1447.20	1942.99	2589.63
kin8nm (RMSE)	0.249	0.250	**0.248**	0.249	0.315
openml-44970 (RMSE)	**1.132**	1.135	1.140	1.190	1.605
space_ga (RMSE)	**0.185**	0.211	0.246	0.208	0.246

Table 8: Random split with offset (block_shift, rate 0.5).

dataset	TabICL-M	TabPFN-3	TabPFN 2.5	TabICLv2	CatBoost
adult	**0.860**	0.858	0.859	0.859	0.843
bank-marketing	0.819	0.818	**0.820**	0.816	0.794
breast cancer	0.994	**0.995**	0.995	0.994	0.990
churn	0.818	**0.822**	0.819	0.813	0.791
climate-model	0.880	**0.881**	0.865	0.866	0.787
cmc	0.679	0.678	**0.681**	0.675	0.650
credit-g	0.725	0.720	**0.727**	0.727	0.695
ilpd	0.708	**0.710**	0.708	0.708	0.677
kc1	0.783	0.780	**0.783**	0.779	0.755
kc2	**0.836**	0.832	0.833	0.828	0.764
pima-diabetes	**0.738**	0.731	0.738	0.730	0.712
qsar-biodeg	**0.913**	0.909	0.909	0.910	0.888
wine	0.988	0.986	0.989	**0.989**	0.981
Boston housing (RMSE)	**4.616**	4.632	4.667	4.700	5.077
bodyfat (RMSE)	3.738	**3.636**	3.738	3.890	4.016
diabetes (RMSE)	64.15	**63.72**	64.11	64.78	69.41
diamonds (RMSE)	1114.00	**1091.35**	1096.38	1098.75	1225.92
kin8nm (RMSE)	0.221	**0.219**	0.220	0.221	0.226
openml-44970 (RMSE)	1.102	1.102	1.101	**1.101**	1.131
space_ga (RMSE)	0.169	**0.166**	0.167	0.168	0.172

Table 9: Complete data (no deleted cells), random split.

dataset	TabICL-M	TabPFN-3	TabPFN 2.5	TabICLv2	CatBoost
adult	0.913	0.911	**0.915**	0.912	0.896
bank-marketing	0.928	0.932	**0.934**	0.931	0.911
breast cancer	**0.997**	0.997	0.996	0.996	0.996
churn	0.929	**0.937**	0.930	0.934	0.927
climate-model	**0.964**	0.961	0.960	0.961	0.947
cmc	0.752	**0.758**	0.757	0.757	0.734
credit-g	**0.817**	0.816	0.815	0.814	0.802
ilpd	0.757	0.760	0.749	**0.772**	0.728
kc1	0.844	0.842	0.833	**0.847**	0.801
kc2	0.835	0.841	**0.845**	0.843	0.783
pima-diabetes	0.846	0.846	0.846	**0.848**	0.826
qsar-biodeg	0.943	0.942	0.942	**0.945**	0.930
wine	**1.000**	1.000	1.000	1.000	0.998
Boston housing (RMSE)	**2.697**	2.709	2.715	2.712	2.817
bodyfat (RMSE)	1.622	1.788	1.939	1.787	**1.389**
diabetes (RMSE)	56.02	56.31	56.33	**55.79**	59.55
diamonds (RMSE)	621.50	**591.35**	609.34	618.61	699.49
kin8nm (RMSE)	0.076	**0.071**	0.077	0.077	0.120
openml-44970 (RMSE)	0.840	0.841	0.843	**0.833**	0.860
space_ga (RMSE)	0.108	**0.094**	0.095	0.102	0.123

C Training hyperparameters

Table 10: The two stages that produced the shipped checkpoints. Both start from the previous weights with a fresh cosine schedule (5 % warm-up).

	stage 4	stage 4b
steps	10 000	10 000
learning rate	5×10^{-5}	3×10^{-5}
batch (tables) / micro-batch	32 / 1	32 / 1
rows per table	400–8 192, log-uniform	400–8 192, log-uniform
features per table	1–100	1–100
optimiser	Muon (β_1 0.9, wd 0.01, clip 1.0)	same
precision	bfloat16 autocast	bfloat16 autocast
reconstruction	mode mixed, weight 0.1, up to 30 % of cells, $p = 0.5$	same
offset consistency	weight 0.1, $p = 0.25$, tables $\leq 2\,048$ rows	same, shift $\leq 1.2\sigma$
prior: tables with gaps	70 %	70 %
prior: block / cell-wise	90 % / 40 %	90 % / 40 %
prior: sources	2–8, Dirichlet shares	2–8
prior: features observed per source	30–100 %	20–100 %
prior: contiguous sources (test-only source)	75 %	90 %
prior: per-source offset	$p = 0.8$, up to 0.5σ	$p = 1.0$, up to 1.2σ
prior: per-source noise	$p = 0.6$, up to 0.3σ	$p = 0.8$, up to 0.5σ
prior: source-id column	$p = 0.5$	$p = 0.5$
hardware, time per task	1 RTX 5090, 9.5 h	1 RTX 5090, 9.5 h

D Engineering notes

Two details mattered in practice. Under distributed data parallel, an out-of-memory micro-batch that was skipped while gradient synchronisation was armed left the reducer waiting for a backward pass that never came; gradients are now averaged by hand after the micro-batch loop, which makes skipped micro-batches and parameters unused in a micro-batch harmless. And the zero-initialised parameters must still take part in every backward pass, so the all-false mask is kept in training mode and an empty reconstruction loss is written as $0 \cdot \sum \theta_{\text{head}}$.

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